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Corpus-Adaptive Symptom Prototype Learning for Social Media Depression Detection

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Abstract

Psychiatric-scale-guided screening has shown promise for social-media depression detection, but existing methods largely treat clinical symptom templates as fixed inputs and focus on improving screening or classification architectures. We instead investigate whether the symptom representation itself can be adapted to social-media language while retaining psychiatric priors. Fixed clinical templates exhibit three key mismatches with online expression: formal versus colloquial language, limited coverage of platform-specific signals, and linguistic distance that can weaken semantic matching. To address these mismatches, we propose Corpus-Adaptive Symptom Prototype Learning, which derives complementary symptom prototypes from depressed users' training posts through semantic clustering and symptom extraction, refinement and deduplication, and social-media-style rewriting. The learned prototypes are combined with existing psychiatric templates and integrated in a plug-and-play manner without modifying downstream detection architectures. Experiments on eRisk2017 and eRisk2018 across five structurally diverse detection backbones show that the proposed representation consistently improves F1 and Recall across all ten dataset–backbone combinations. Ablation results further show that the gains arise from corpus-derived symptom information rather than LLM-generated reformulations alone, and that all three prototype-construction stages contribute to performance. These results demonstrate that adapting symptom representations, rather than only detection architectures, provides an effective and architecture-agnostic extension to psychiatric-scale-guided screening. Code, pre-trained model, and data processing workflow are publicly available at https://github.com/GreenProgramm/CASP.

Keywords: depression detection, social media, adaptive corpus, symptom prototype, psychiatric scale

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1. Introduction

Depression remains a major global health challenge, while high treatment costs, persistent stigma, and limited mental-health infrastructure prevent many individuals from receiving timely support. Social media provides a complementary behavioral record in which users spontaneously describe emotional states, daily struggles, and psychological experiences. Computational analysis of these records therefore offers a scalable means of identifying potentially at-risk users and complementing traditional diagnostic channels [1–3].

A widely adopted paradigm for social-media depression detection separates the task into two stages: a screening module first selects a compact set of high-risk posts from a user's long posting history, after which a detection model performs user-level classification. Zhang et al. [4] introduced psychiatric-scale-guided screening using symptom templates derived from the Beck Depression Inventory-II (BDI-II) [5], while Wang et al. [6] further developed end-to-end Learnable Psychiatric Scale-guided risky post screening (E2-LPS) to jointly optimize screening and detection. Both confirm the value of psychiatric guidance for post screening.

However, existing work largely treats psychiatric symptom templates as fixed clinical inputs and concentrates on improving screening or classification architectures. In contrast, *we shift the focus from architectural design to the symptom representation used for screening*, investigating how fixed clinical knowledge can be adapted to better reflect the linguistic characteristics of social media while preserving psychiatric priors. Our analysis of existing clinical templates reveals three major mismatches with social-media language: **(i)** clinical descriptions are relatively abstract and formal, whereas users often express distress through colloquial, fragmented, or implicit language; **(ii)** fixed clinical dimensions may not fully capture platform-specific manifestations of depression that emerge from online discourse [7]; and **(iii)** the linguistic gap between clinical descriptions and social-media posts can weaken semantic matching, causing implicit depression-related expressions to be overlooked.

To address this representation mismatch, we propose a corpus-adaptive symptom prototype learning framework that constructs complementary symptom prototypes directly from depressed users' training posts while preserving the original psychiatric templates. Rather than introducing a new depression-classification architecture, our method adapts the screening representation itself. The framework contains three stages. First, semantic clustering partitions depressed users' training posts into recurring expression patterns. Second, GPT-5.5 Instant [8] extracts structured symptom specifications from these clusters and refines them by merging redundant or overlapping patterns. Third, the refined specifications are rewritten into natural social-media-style descriptions suitable for semantic similarity matching. The resulting corpus-derived prototypes are combined with the original psychiatric templates to form a richer screening representation. Importantly, the proposed prototypes are architecture-agnostic and plug-and-play: they can replace or augment the fixed screening templates without modifying the downstream detection architecture. This isolates the contribution of symptom representation quality.

Across BERT [9], HAN-BERT [4], E2-LPS [6], LLaMA-3.1 [10], and Qwen-3 [11] on both eRisk2017 [12] and eRisk2018 [13], incorporating the learned prototypes consistently improves F1 and Recall under otherwise unchanged backbone-specific protocols.

Our main contributions are summarized as follows:

- We reformulate psychiatric-scale-guided depression screening from an architecture-centered problem to a *symptom-representation adaptation* problem, and reveal three mismatches between fixed clinical templates and social-media language: expression style, coverage of platform-specific signals, and semantic matching distance.
- We propose a three-stage corpus-adaptive pipeline that learns complementary symptom prototypes from depressed users' training posts while retaining the original psychiatric templates, thereby combining clinical priors with corpus-specific depression-related expressions.
- We introduce an architecture-agnostic, plug-and-play integration strategy that improves F1 and Recall across five structurally diverse detection backbones on both eRisk2017 and eRisk2018 without modifying the downstream detection architectures.

2. Related Work

2.1. Depression Detection on Social Media

Computational approaches to depression detection have been studied extensively since the introduction of dedicated benchmarks and shared tasks [12–14]. Early feature-based methods extracted linguistic signals such as TF-IDF weights, LIWC psycholinguistic categories, and behavioral posting patterns [15,16]. Deep learning models subsequently improved performance through hierarchical content modeling [2,17,18], while pretrained language models such as BERT have advanced benchmark results further. More recently, Yang et al. combined transfer learning with multi-source feature fusion and imbalance mitigation for social-media mental-health detection, improving BERT- and RoBERTa-based classification performance [19]. A persistent challenge is the high volume of posts per user, most of which carry no depression signal. To address this, Zogan et al. [20] proposed selecting representative posts via K-means clustering on Sentence-BERT embeddings [21] and compressing them through abstractive summarisation with BART [22]. However, compression-based approaches may lose fine-grained symptom signals scattered across individual posts. Our method instead constructs corpus-adaptive symptom prototypes directly from depressed users' posts, enabling more targeted screening without post compression.

2.2. Psychiatric Scale-Guided Screening

Zhang et al. [4] proposed the first psychiatric scale-guided post-screening method, constructing depression templates from the BDI-II [5] to filter risky posts via cosine similarity, then classifying users with a hierarchical attention network incorporating BERT (HAN-BERT). A recent study in *Electronics* integrated DSM-5-guided textual, behavioral, and topic features with hierarchical BERT-based multi-task learning, further illustrating the value of explicit clinical knowledge for interpretable social-media depression screening [23]. Wang et al. extended this paradigm with E2-LPS [6], enabling end-to-end joint training of the screening and detection modules via straight-through estimators. Both works demonstrate the critical role of psychiatric priors in screening, yet neither examines whether the clinical templates are well-matched to the social media domain. Our work addresses this gap by augmenting $\mathcal{T}_{\text{psych}}$ with corpus-adaptive symptom prototypes grounded in actual user expression.

2.3. Large Language Models for Mental Health

Large language models have shown promise for mental-health analysis by supporting flexible text interpretation, explainable reasoning, and longitudinal modeling [24–26]. Recent studies from NLPCC 2025 further demonstrate the growing role of LLMs in mental health and affective computing [27,28]. Lan et al. proposed DORIS [25], a hybrid framework that uses LLMs to annotate depression-related symptoms according to DSM-5 criteria [29] and to summarize longitudinal mood trajectories from users' historical posts. Rather than relying on the LLM as the final classifier, DORIS combines these clinically informed representations with post-history features and feeds them into a Gradient Boosting Tree classifier, while the LLM further generates evidence-supported explanations from the identified symptoms and mood-course patterns. This design highlights the potential of LLMs as clinically grounded feature-construction and interpretation components rather than solely as end-to-end predictors. However, deploying proprietary LLMs as end-to-end classifiers raises concerns about reproducibility, inference cost, and data privacy. In our work, LLMs serve two distinct roles: GPT-5.5 [8] is used offline as a one-time knowledge extractor during prototype construction, with no test-set data involved; LLaMA-3.1 [10]

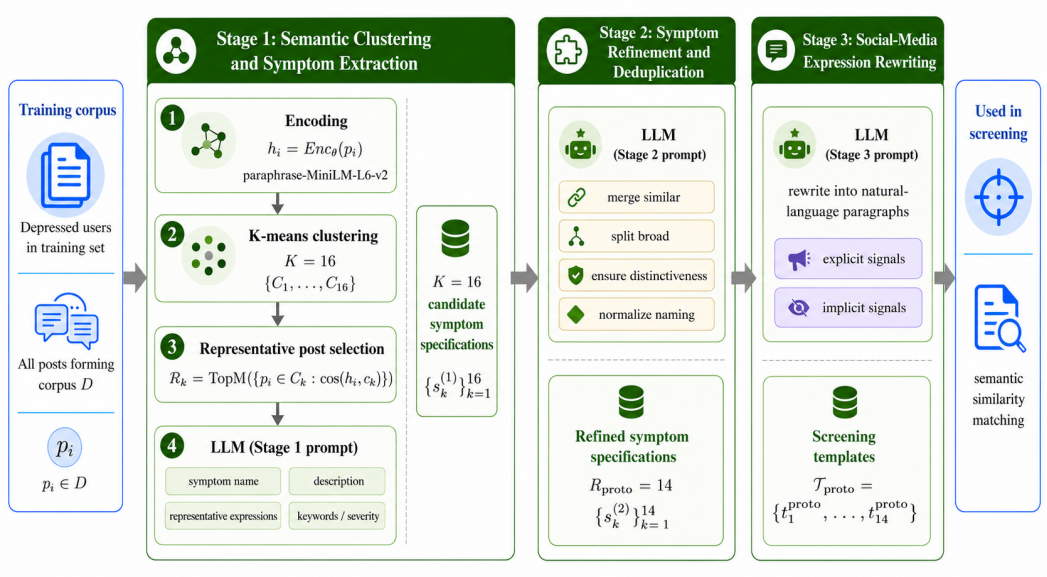


Figure 1. Training-only three-stage pipeline for corpus-adaptive symptom prototype learning.

and Qwen-3 [11] serve as open-source classification baselines deployable locally, avoiding API dependency and ensuring reproducibility.

3. Methodology

3.1. Problem Formulation

Given a user U with historical posts $\mathcal{P}_U = \{p_1, \ldots, p_n\}$, the goal is to predict a binary dataset label $y \in \{0, 1\}$, where $y = 1$ denotes the depressed class. Under the screening–detection paradigm, each post is first assigned a risk score using a set of symptom templates, and a backbone-specific screening budget selects the highest-risk posts for user-level classification.

Let $\mathcal{T} = \{t_j\}_{j=1}^{|\mathcal{T}|}$ denote the screening template set. A screening encoder maps post p_i and template t_j to embeddings $\mathbf{h}_i$ and $\mathbf{z}_j$, respectively. The risk score of p_i is

$$r_i(\mathcal{T}) = \max_{t_j \in \mathcal{T}} \cos(\mathbf{h}_i, \mathbf{z}_j) = \max_{t_j \in \mathcal{T}} \frac{\mathbf{h}_i^\top \mathbf{z}_j}{\|\mathbf{h}_i\| \, \|\mathbf{z}_j\|}. \tag{1}$$

Prior psychiatric-scale-guided screening uses 24 fixed templates $\mathcal{T}_{\mathrm{psych}}$ [6]. Rather than modifying the downstream detector, we adapt the *screening representation* itself by learning an additional corpus-derived template set $\mathcal{T}_{\mathrm{proto}}$. The two sources of symptom knowledge are combined as

$$\mathcal{T}_{\mathrm{comb}} = \mathcal{T}_{\mathrm{psych}} \cup \mathcal{T}_{\mathrm{proto}}, \qquad |\mathcal{T}_{\mathrm{comb}}| = 24 + 14 = 38. \tag{2}$$

Thus, the screening rule and downstream architecture remain unchanged; only the template representation used to compute r_i is augmented.

3.2. Corpus-Adaptive Symptom Prototype Learning

Fig. 1 illustrates the three-stage construction pipeline. For each dataset, prototypes are constructed exclusively from its training split; the official test set is never accessed during prototype construction.

3.2.1. Stage 1: Semantic Clustering and Symptom Extraction

We collect all posts from users labeled as depressed in the training set, forming corpus $\mathcal{D}$. Each post $p_i \in \mathcal{D}$ is encoded using paraphrase-MiniLM-L6-v2 [21] as $\mathbf{e}_i = \mathrm{Enc}_\theta(p_i)$. K-means partitions the embeddings into $\{\mathcal{C}_1, \ldots, \mathcal{C}_K\}$. For each cluster $\mathcal{C}_k$ with centroid $\mathbf{c}_k$, we select the Top-M posts with the highest cosine similarity to the centroid:

$$\mathcal{R}_k = \mathrm{TopM}_{p_i \in \mathcal{C}_k} \cos(\mathbf{e}_i, \mathbf{c}_k). \tag{3}$$

The representative set $\mathcal{R}_k$ is then provided to GPT-5.5 [8] with the Stage 1 prompt, which identifies one shared depression-related psychological or expression pattern and returns a structured specification containing a symptom name, description, representative expressions, keywords, and severity. This yields $\mathcal{S}^{(1)} = \{s_k^{(1)}\}_{k=1}^{K}$.

We set $K = 16$ to balance semantic granularity and candidate redundancy, and Top-$M = 16$ to balance within-cluster expression coverage with LLM context length and computational cost. Both are fixed before official test evaluation and shared across datasets.

3.2.2. Stage 2: Symptom Refinement and Deduplication

Because independently extracted clusters may describe overlapping psychological patterns, all Stage 1 candidates are jointly provided to GPT-5.5 with the Stage 2 prompt. The model merges semantically similar candidates, splits overly broad patterns when necessary, removes redundant or overly generic entries, normalizes naming, and preserves distinct corpus-specific expressions. The resulting set contains $R_{\mathrm{proto}} = 14$ refined specifications,

$$\mathcal{S}^{(2)} = \{s_k^{(2)}\}_{k=1}^{R_{\mathrm{proto}}}, \qquad R_{\mathrm{proto}} = 14. \tag{4}$$

3.2.3. Stage 3: Social-Media Expression Rewriting

The refined specifications remain relatively structured and formal, which can create a linguistic gap with informal social-media posts. For each $s_k^{(2)}$, the Stage 3 prompt instructs GPT-5.5 to integrate its symptom description, representative expressions, and keywords into a natural social-media-style paragraph that captures both explicit and implicit manifestations. This produces one screening template per refined specification,

$$\mathcal{T}_{\mathrm{proto}} = \{t_1^{\mathrm{proto}}, \ldots, t_{R_{\mathrm{proto}}}^{\mathrm{proto}}\}, \qquad |\mathcal{T}_{\mathrm{proto}}| = 14. \tag{5}$$

The resulting paragraphs are used directly for embedding-based semantic matching. The complete Stage 1–3 prompts and an illustrative cluster-to-prototype trace are provided in Appendix A.

3.3. Architecture-Agnostic Plug-and-Play Integration

Once constructed, $\mathcal{T}_{\mathrm{proto}}$ augments the fixed psychiatric templates through $\mathcal{T}_{\mathrm{comb}}$ without changing the downstream detection architecture or its original training protocol. For static screening with BERT and HAN-BERT, $\mathcal{T}_{\mathrm{comb}}$ replaces the original template bank when computing the post risk scores in (1), after which the backbone-specific screening budget selects posts for classification. For E2-LPS, the same combined template set is incorporated into its learnable screening module while retaining the original detection architecture. For LLM backbones, $\mathcal{T}_{\mathrm{comb}}$-screened posts use the same inference protocol as the baseline.

This plug-and-play design keeps the downstream backbone fixed in each paired comparison, allowing the effect of corpus-adaptive symptom representations to be evaluated independently of architectural changes.

4. Experiments

4.1. Datasets and Experimental Setup

4.1.1. Datasets

We conduct experiments on two widely-used benchmarks: eRisk2017 [12] and eRisk2018 [13], both collected from Reddit. Table 1 summarises dataset statistics.

Table 1. Dataset statistics.

Dataset		No. of Users		No. of Posts	
		Depressed	Control	Depressed	Control
Train	eRisk2017	83	403	30,851	264,172
	eRisk2018	135	752	49,557	481,837
Test	eRisk2017	52	349	18,706	217,665
	eRisk2018	79	741	40,665	504,523

Table 2. Main results of five detection backbones on eRisk2017 and eRisk2018 datasets.

Method	eRisk2017					eRisk2018				
	Precision	Recall	F1	ROC-AUC	PR-AUC	Precision	Recall	F1	ROC-AUC	PR-AUC
BERT (Clus)	0.4821	0.5192	0.5000	0.7907	0.4160	0.4557	0.4557	0.4557	*0.8270*	0.3857
BERT (Clus+Abs)	0.4138	0.4615	0.4364	0.6873	0.3341	0.4627	0.3924	0.4247	0.6812	0.3298
BERT (Psych)	**0.5686**	*0.5577*	*0.5631*	*0.8466*	*0.5569*	**0.5658**	*0.5443*	*0.5548*	0.8247	*0.4793*
BERT (Ours)	*0.5536*	**0.5962**	**0.5741**	**0.8508**	**0.5652**	*0.5495*	**0.6329**	**0.5882**	**0.8340**	**0.5279**
HAN-BERT (Clus)	0.5789	*0.6346*	0.6055	0.8701	0.5317	0.4696	*0.6835*	0.5567	*0.9119*	0.6201
HAN-BERT (Clus+Abs)	0.4038	0.4038	0.4038	0.7096	0.3695	0.5532	0.3291	0.4127	0.7761	0.3303
HAN-BERT (Psych)	**0.8000**	0.6154	*0.6957*	*0.8766*	*0.6508*	**0.6500**	0.6582	*0.6541*	0.9118	*0.6475*
HAN-BERT (Ours)	*0.7170*	**0.7308**	**0.7238**	**0.8809**	**0.6932**	*0.6100*	**0.7722**	**0.6816**	**0.9263**	**0.6661**
E2-LPS	**0.8333**	*0.5769*	*0.6818*	*0.8956*	*0.7187*	**0.6512**	*0.7089*	*0.6788*	*0.9425*	*0.7057*
E2-LPS (Ours)	*0.7600*	**0.7308**	**0.7451**	**0.9257**	**0.7607**	*0.6477*	**0.7215**	**0.6826**	**0.9443**	**0.7101**
LLaMA-3.1-8B (Psych)	0.2193	*0.7885*	0.3431	–	–	0.1528	*0.7215*	0.2522	–	–
LLaMA-3.1-8B (E2-LPS)	*0.2390*	0.7308	*0.3602*	–	–	*0.1599*	0.6456	*0.2563*	–	–
LLaMA-3.1-8B (Ours)	**0.2471**	**0.8269**	**0.3805**	–	–	**0.1695**	**0.7595**	**0.2771**	–	–
Qwen-3-8B (Psych)	0.2901	*0.7308*	0.4153	–	–	*0.2460*	*0.7722*	*0.3731*	–	–
Qwen-3-8B (E2-LPS)	**0.3596**	0.6154	*0.4539*	–	–	0.2295	0.5316	0.3206	–	–
Qwen-3-8B (Ours)	*0.3421*	**0.7500**	**0.4699**	–	–	**0.2520**	**0.8101**	**0.3844**	–	–

Note: Bold/italic denote the best/second-best result within each backbone group; – indicates not applicable.

4.1.2. Evaluation metrics

Following prior work on social-media depression detection [25], we evaluate the models using Precision, Recall, F1-score, area under the ROC curve (ROC-AUC), and area under the precision–recall curve (PR-AUC), with depressed users treated as the positive class. Precision measures the reliability of positive depression predictions, whereas Recall reflects the ability to identify users at risk and is particularly relevant in screening scenarios where missed positive cases are undesirable. F1-score summarizes the trade-off between Precision and Recall. ROC-AUC evaluates the model's overall ranking ability across decision thresholds. Since depression-detection datasets are typically class-imbalanced, PR-AUC is especially informative because it focuses on performance for the positive class and is therefore treated as a primary threshold-independent metric.

4.1.3. Screening strategies

Clus applies K-means clustering to post embeddings without psychiatric-scale guidance. **Clus+Abs** further summarizes the resulting clusters using BART [22]. **Psych** performs static screening using the 24 psychiatric-scale templates $\mathcal{T}_{\text{psych}}$ adopted from prior work [4].

E2-LPS uses end-to-end learnable screening with $\mathcal{T}_{\text{psych}}$ [6]. **Ours** uses the combined template set $\mathcal{T}_{\text{comb}} = \mathcal{T}_{\text{psych}} \cup \mathcal{T}_{\text{proto}}$, containing 38 templates in total.

4.1.4. Hardware and reproducibility

All experiments are implemented with PyTorch 2.2 and conducted on two NVIDIA RTX 3090 GPUs. Following prior work [4,6], we run each model with three seeds and report the best result.

4.2. Implementation Details of Detection Backbones

4.2.1. BERT and HAN-BERT

We use BERT-base-uncased as the backbone encoder. For post screening, $\text{Enc}_\theta(\cdot)$ is instantiated as paraphrase-MiniLM-L6-v2 [21], and the number of screened posts is set to 16 following [4]. The selected posts are concatenated for BERT, while HAN-BERT uses a 4-layer Transformer user encoder with 8 attention heads. The batch size is 4 and the learning rate is 2×10^{-5}.

4.2.2. E2-LPS

We use all-MiniLM-L6-v2 as the screening encoder and BERT-base-uncased as the detection backbone. Following [6], the screening ratio is 12.5%. The batch size is 1, with learning rates of 1×10^{-5} for BERT and 2×10^{-5} for the remaining trainable components. The user encoder is a single-layer Transformer with 8 attention heads. All neural models are optimized with AdamW for 10 epochs.

4.2.3. LLM baselines

LLaMA-3.1-8B [10] and Qwen-3-8B [11] are deployed with vLLM. We use balanced 2-shot prompting with one depressed and one control example randomly sampled from the training set for each test instance. Each user input is limited to 12,000 characters, with temperature $= 0$, top-$p = 1$, and at most 256 generated tokens. The model outputs a single label, `depressed` or `control`. Inference uses tensor parallelism of size 2, GPU memory utilization of 0.90, and a maximum model length of 32,768 tokens. Because this protocol produces hard labels rather than continuous prediction scores, ROC-AUC and PR-AUC are not applicable to the two LLM backbones.

4.3. Main Results and Analysis

Table 2 reports the results on eRisk2017 and eRisk2018 under all five evaluation metrics.

4.3.1. Consistent gains from symptom prototypes

Replacing the corresponding baseline screening strategy with *Ours* improves both F1 and Recall across all five backbone types on both datasets. For example, HAN-BERT improves from 0.6957 to 0.7238 in F1 and from 0.6154 to 0.7308 in Recall on eRisk2017; on eRisk2018, the corresponding improvements are from 0.6541 to 0.6816 and from 0.6582 to 0.7722. These consistent gains indicate that the corpus-derived prototypes capture depression-related expressions complementary to those represented by the fixed psychiatric templates.

4.3.2. Threshold-independent gains and operating-point trade-offs

For BERT, HAN-BERT, and E2-LPS, *Ours* improves ROC-AUC and PR-AUC in all six dataset–backbone comparisons, demonstrating stronger threshold-independent discrimination. At the selected decision threshold, these gains are accompanied by higher Recall but lower Precision than the corresponding baseline in all six comparisons. This reflects a

recall-oriented operating-point trade-off: more depressed users are identified, at the cost of additional false positives. Such behavior is particularly relevant in early-risk screening, where missed potentially at-risk users may be more consequential than additional candidates passed to subsequent assessment.

4.3.3. Generalization across backbone architectures

The improvements persist across standard BERT, hierarchical HAN-BERT, the end-to-end E2-LPS framework, and decoder-only LLMs, suggesting that the benefit of the proposed prototypes is not tied to a particular classifier architecture. Moreover, the weaker performance of *Clus* and *Clus+Abs* relative to psychiatric-guided variants supports the value of retaining psychiatric priors. Accordingly, our method augments rather than replaces $\mathcal{T}_{\text{psych}}$ with corpus-adaptive information.

4.3.4. LLM backbones

LLaMA-3.1 and Qwen-3 exhibit relatively high Recall but lower Precision and absolute F1, indicating a tendency toward positive predictions. This behavior is consistent with recent findings in social-media depression detection. Lan et al. reported that direct classification with GPT-4o mini and MentalLLaMA produced high Recall but very low Precision and F1, and similarly concluded that directly using LLMs for depression classification can lead to poor classification performance [25]. These models are therefore included here not as proposed depression classifiers, but as structurally different backbones for testing whether the proposed prototypes remain useful beyond encoder-based architectures. The relevant comparison is therefore between *Psych* and *Ours* under the same LLM inference protocol. Across both LLMs and both datasets, *Ours* improves Recall and F1 in all four paired comparisons. ROC-AUC and PR-AUC are not reported for the LLM backbones because the current prompting protocol produces only hard labels rather than continuous decision scores.

4.4. Ablation Study

All ablation experiments are conducted on eRisk2017 with HAN-BERT [4] as the detection backbone.

4.4.1. Effect of template combination

As shown in Table 3, Psych-only uses the 24 fixed psychiatric-scale templates from prior work, whereas Proto-only is obtained through corpus-driven extraction, refinement, and social-media-style rewriting. Because Stage 3 rewrites the intermediate symptom specifications produced by Stages 1–2, the fixed Psych templates cannot undergo the same pipeline directly. To control for LLM-generated wording, we construct Psych Expansion by using GPT-5.5 to generate five additional social-media expressions for each Psych template, yielding 144 templates without accessing eRisk posts.

Psych Expansion raises Recall from 0.6154 to 0.6923 but lowers F1 from 0.6957 to 0.6545. Proto-only achieves higher Recall (0.7500) with comparable F1 (0.6964), while combining Psych and Proto gives the best F1 (0.7238) and Recall of 0.7308. Thus, the gains cannot be attributed to LLM-generated reformulations alone; corpus-derived prototypes provide complementary information to the fixed psychiatric templates.

Table 3. Effect of template combination on eRisk2017.

Method	Recall	F1
HAN-BERT (Psych-only)	0.6154	0.6957
HAN-BERT (Psych Expansion)	0.6923	0.6545
HAN-BERT (Proto-only)	**0.7500**	0.6964
HAN-BERT (Psych+Proto, Ours)	0.7308	**0.7238**

4.4.2. Stage-wise ablation

In Table 4, S1, S2, and S3 denote three stages: clustering, refinement, and social-media rewriting, respectively. Removing Stage 1 clustering causes the largest degradation, reducing Recall from 0.7308 to 0.5769 and F1 from 0.7238 to 0.6522. Without semantic partitioning, the LLM must infer symptom patterns from the corpus without cluster-level structure, resulting in coarser and less discriminative prototypes. Removing Stage 2 refinement also degrades both metrics, supporting the importance of merging overlapping candidates and filtering redundant or overly generic symptom specifications. Removing Stage 3 rewriting produces the smallest performance drop, but still reduces Recall to 0.6909 and F1 to 0.7103, indicating that adapting the refined specifications to natural social-media language improves semantic matching between posts and prototypes. Overall, all three stages contribute to the final performance, with clustering providing the largest gain.

Table 4. Stage-wise Ablation on eRisk2017.

Variant	S1	S2	S3	Recall	F1
w/o Clustering	×	✓	✓	0.5769	0.6522
w/o Refinement	✓	×	✓	0.6552	0.6909
w/o Rewriting	✓	✓	×	0.6909	0.7103
Full Pipeline	✓	✓	✓	**0.7308**	**0.7238**

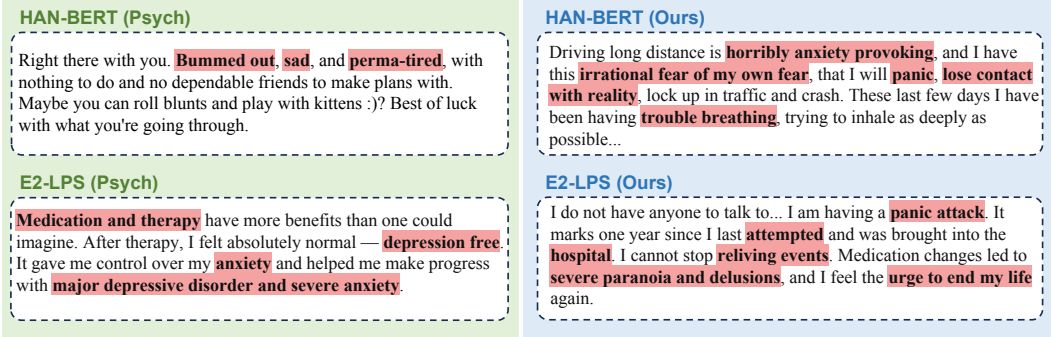


Figure 2. Illustrative screening cases for depressed users from the eRisk2017 dataset.

4.5. Case Study: Qualitative Analysis of Screened Posts

Fig. 2 presents illustrative screening cases for depressed users from the eRisk2017 dataset, with relevant mental health cues highlighted. Compared with $\mathcal{T}_{\text{psych}}$-based screening, the posts selected by our corpus-adaptive symptom prototype learning provide richer insights into users' psychological states, capturing nuanced and platform-specific depressive signals beyond surface-level clinical vocabulary, thereby demonstrating its effectiveness in risky post screening.

4.6. Psychiatric Scale Templates vs. Symptom Prototypes

Table 5 presents the 24 psychiatric-scale templates $\mathcal{T}_{\text{psych}}$ used in prior psychiatric-scale-guided screening [4,6], while Table 6 lists the 14 corpus-adaptive prototype names in $\mathcal{T}_{\text{proto}}$ learned from eRisk2017 or eRisk2018 via our method. Comparing the psychiatric scale templates $\mathcal{T}_{\text{psych}}$ with the two sets reveals how $\mathcal{T}_{\text{proto}}$ addresses the three mismatches: **(i) Abstract vs. colloquial:** BDI-II uses formal declarations (*"I feel worthless"*); prototypes such as *Surface-level social masking* and *Dismissive disengagement* capture how the same states surface in fragmented, informal posts. **(ii) Fixed vs. platform-specific signals:** BDI-II omits online behavior entirely, whereas prototypes such as *Digital disengagement and online cynicism* and *Online belonging insecurity* encode signals unique to social media. **(iii) Linguistic distance:** Stage 3 rewrites prototypes in social-media style, reducing the style gap and cosine-matching noise.

Table 5. Psychiatric-scale templates $\mathcal{T}_{\text{psych}}$ used for screening.

No.	Statement
1	I feel depressed.
2	I am diagnosed with depression.
3	I am treating my depression.
4	I feel sad.
5	I am discouraged about my future.
6	I always fail.
7	I don't get pleasure from things.
8	I feel quite guilty.
9	I expected to be punished.
10	I am disappointed in myself.
11	I always criticize myself for my faults.
12	I have thoughts of killing myself.
13	I always cry.
14	I am hard to stay still.
15	It's hard to get interested in things.
16	I have trouble making decisions.
17	I feel worthless.
18	I don't have energy to do things.
19	I have changes in my sleeping pattern.
20	I am always irritable.
21	I have changes in my appetite.
22	I feel hard to concentrate on things.
23	I am too tired to do things.
24	I have lost my interest in sex.

Table 6. Symptom Prototypes Learned from eRisk2017 and eRisk2018 Training Sets.

No.	eRisk2017	eRisk2018
1	Digital disengagement and online cynicism	Eating-related preoccupation and control conflict
2	Existential worthlessness and nihilism	Appearance-based self-scrutiny
3	Existential insignificance	Romantic rejection sensitivity and intimacy deprivation
4	Emotional blunting	Attachment hurt and relational ambivalence
5	Dismissive disengagement	Interpersonal disillusionment and betrayal sensitivity
6	Disordered eating preoccupation	Online belonging insecurity
7	Entrapment and perceived stagnation	Support-seeking and social appeasement
8	Paranoid ideation	Vocational stagnation and directionlessness
9	Indecisiveness and doubt	Crisis-level emotional overwhelm
10	Cynical distrust in institutions	Chronic pain entrapment and medical invalidation
11	Self-critical rumination	Emotional blunting and low engagement
12	Surface-level social masking	Escapist emotional immersion
13	Interpersonal cynicism and distrust	Reactive irritability and defensive anger
14	Fearful attachment	Externalized frustration and control-seeking

The existing 24 psychiatric-scale templates retain the clinical foundation of the screening system; the learned prototypes are complementary social-media expression patterns intended to reduce the mismatch between fixed clinical wording and corpus language, rather than new diagnostic constructs. After construction, the prototypes were informally reviewed by a clinician experienced in treating depression and were judged broadly plausible. For example, "Self-critical rumination" closely corresponds to depression-related guilt, worthlessness, and self-criticalness. "Emotional blunting and low engagement" relates to diminished interest, reduced activity, and low energy. By contrast, "Digital disengagement and online cynicism" has only an indirect association with diminished interest and reduced engagement; its digital form is platform-specific. "Cynical distrust in institutions" has no direct DSM-5 or BDI-II counterpart and may capture a corpus-specific negative worldview or topic. It should therefore not be interpreted as a standalone clinical symptom.

5. Conclusions

We proposed a Corpus-Adaptive Symptom Prototype Learning framework for social-media depression detection that complements fixed psychiatric templates with corpus-derived prototypes reflecting informal, heterogeneous, and platform-specific expressions of depressive symptoms. The proposed prototypes are integrated in a plug-and-play manner without modifying downstream detection architectures. Experiments on both eRisk2017 and eRisk2018 show that the learned prototypes consistently improve Recall and F1 across encoder-based, end-to-end, and LLM backbones, supporting the effectiveness and architecture-agnostic nature of the proposed adaptation strategy.

The framework is intended solely as a computational research tool for depression-risk screening rather than as a clinical diagnostic system. Although downstream evaluation on both depressed and control users demonstrates the computational usefulness of the learned prototypes, it does not establish their clinical specificity. Future work will investigate contrastive prototype construction using both positive and control users, including explicit mappings that distinguish direct clinical associations, indirect associations, and corpus-specific patterns. We also plan to conduct structured evaluations with multiple clinical experts to further assess the plausibility, consistency, and clinical relevance of the learned prototypes.

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Data Availability Statement: The code, pre-trained model, and data processing workflow are publicly available at https://github.com/GreenProgramm/CASP. Cluster inputs will be shared where permitted by the dataset licenses. The eRisk2017 and eRisk2018 datasets are not redistributed with this article and remain subject to the original access and redistribution terms of the dataset providers.

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Appendix A. Complete Prompts and Illustrative Cluster-to-Prototype Trace

Appendix A.1. Stage 1 Prompt: Symptom Extraction—Complete Verbatim Prompt

You are a clinical psychologist specializing in depression-related language analysis.

You are given a group of semantically similar social-media posts written by users labeled as depressed in the training corpus. Your task is not to summarize the posts, but to identify the underlying psychological symptom or depression-related expression pattern shared by them.

Requirements:

1. Identify one core symptom or psychological expression pattern.
2. Use a specific and meaningful label rather than a generic term such as "negative emotion."
3. Consider both explicit and implicit expressions.
4. Generalize from the posts without copying them verbatim.
5. Provide:
   - a short symptom name;
   - a detailed description;
   - five representative expressions;
   - three to five keywords;
   - an estimated severity level: low, medium, or high.

Return the result strictly in the following JSON format:

```
{
 "symptom_name": "...",
 "description": "...",
 "representative_expressions": [
 "...",
 "...",
 "...",
 "...",
 "..."
 ],
 "keywords": [
 "...",
 "...",
 "..."
 ],
 "severity": "low | medium | high"
}
```

Posts:

```
{cluster_posts}
```

Appendix A.2. Stage 2 Prompt: Refinement and Deduplication—Complete Verbatim Prompt

You are refining a set of depression-related symptom prototypes extracted from different semantic clusters.

Given the candidate symptom specifications below, perform the following tasks:

1. Merge symptoms that describe the same or highly similar psychological pattern.
2. Split an overly broad symptom only when it contains clearly distinct patterns.
3. Remove duplicate, overly generic, or non-informative symptoms.

4. Ensure that each retained symptom is semantically distinct and clinically interpretable.
5. Normalize the naming style across all symptoms.
6. Preserve important corpus-specific depression-related expressions when they are not already covered by another prototype.

Return the refined symptom set strictly as a JSON list. Each item must follow this format:

```
[
 {
 "symptom_name": "...",
 "description": "...",
 "representative_expressions": [
 "...",
 "...",
 "...",
 "...",
 "..."
 ],
 "keywords": [
 "...",
 "...",
 "..."
 ],
 "severity": "low | medium | high",
 "merged_from": [
 "..."
 ]
 }
]
```

Candidate symptoms:

```
{candidate_symptoms}
```

Appendix A.3. Stage 3 Prompt: Social-Media-Style Rewriting—Complete Verbatim Prompt

You are a mental-health language expert.

Given a structured description of a depression-related symptom, rewrite it as one natural and coherent paragraph reflecting how people may express this experience on social media.

Requirements:

1. Integrate the symptom name, description, representative expressions, and keywords into a single paragraph.
2. Use natural and informal language similar to real social-media posts.
3. Include both explicit and implicit manifestations of the symptom.
4. Do not use bullet points, headings, or lists.
5. Do not copy the input expressions verbatim.
6. Avoid diagnostic claims or statements that every person with this experience has depression.
7. Make the paragraph suitable for sentence-embedding-based semantic similarity matching.
8. Output only the rewritten paragraph.

Structured symptom:

```
{symptom_specification}
```

Appendix A.4. Illustrative Cluster-to-Prototype Trace (eRisk2017)

The expressions quoted below are the generated representative expressions contained in the Stage-1 candidate specifications, rather than verbatim reproductions of the raw cluster posts.

One Stage-1 cluster produced the candidate "Entrapment and helplessness," characterized by representative expressions such as "It feels like I'm sinking and nothing I do actually stops it," "No matter what I try, I can't seem to make things better," and "I feel stuck in this state and don't see a real way out." A second semantically related cluster produced "Perceived Life Stagnation," with expressions including "I start doing well for a day, but everything piles up again" and "I want something different in my life, but … end up stuck where I am."

In Stage 2, these overlapping candidates were consolidated into "Entrapment and perceived stagnation":

A pervasive sense of being stuck in an unchanging and unsatisfactory state, combined with helplessness and inability to create meaningful progress. Efforts to improve feel ineffective, leading to reduced agency and resignation.

Stage 3 rewrote this refined specification as the final screening prototype corresponding to No. 7, "Entrapment and perceived stagnation," in the eRisk2017 column of Table 6:

I feel stuck in the same place no matter what I do, like I'm not moving forward at all. Even when I try to change things, it doesn't seem to make a difference, and eventually I just lose the energy to keep trying. It's like being trapped in a situation that won't shift, and over time it just turns into this quiet feeling that nothing I do will actually work.

This trace illustrates how cluster-derived evidence is converted into Stage-1 candidate patterns, how Stage 2 reduces redundancy by merging semantically related candidates into a single refined specification, and how Stage 3 transforms that specification into the informal screening prototype reported as No. 7 in Table 6 for semantic matching.

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